Work-in-Progress: From Generic Chatbots to Teacher-Configurable AI Tutors: The Design and Evaluation of Novedu in Secondary Education

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Abstract. Secondary-school students increasingly use general-purpose chatbots for learning, but these systems are rarely aligned with classroom context, teacher intentions, or a school's privacy requirements. This work-in-progress paper presents *Novedu*, a teacher-configurable AI-tutoring platform developed at HTL Leonding, an Austrian technical upper-secondary school. Our position is that a school cannot realistically align large language model (LLM) tutors by training models, but can align them institutionally by giving teachers ownership of configuration, curricular grounding, deployment, privacy defaults, and evaluation: tutors are assembled from reusable, teacher-authored modules and served through in-region, open-weight models run by a research partner. A requirements survey (n = 318) yields four central requirements (curricular grounding, factual reliability, configurable scaffolding, and aggregated oversight), each traced to a design response; a reproducible evaluation harness gives preliminary one-topic evidence that an in-region open-weight model can be competitive with frontier baselines, while depending strongly on teacher-authored prompt detail. The platform and classroom evaluation remain ongoing.

Keywords: generative AI in education · teacher-configurable AI tutors · pedagogical alignment · in-region open-weight LLMs

1 Introduction

Generative AI has become an everyday tool in secondary education. At our school adoption is near-universal: 94% of students have used an AI tool, 84.5% regularly (Chapter 3). The opportunities and risks are well rehearsed [3]; what remains unresolved for an individual school is not *whether* students use AI, but *how* the institution can channel that use toward learning while meeting its legal and pedagogical obligations.

For an institution, a general-purpose chatbot is a poor default for two distinct reasons: one pedagogical, one institutional. Pedagogically, used as-is a large language model (LLM) tends to provide immediate answers rather than guiding students through problem solving [7], and it knows neither the class's current topic nor the teacher's intentions. Institutionally, sending student data to commercial cloud LLMs raises data-protection obligations under the General Data Protection Regulation (GDPR) and, for some educational uses such as evaluation or steering of learning, governance obligations under the EU AI Act, and it gives teachers no oversight of its use in their subject.

This paper presents *Novedu*, a teacher-configurable AI-tutoring software developed as a cooperative student software-engineering project at HTL Leonding, a school of higher technical vocational education in Austria with a strong informatics profile, and deployed at the school. Our central position is that a school cannot realistically align tutors by *training* models, but it can align them *institutionally*: by giving teachers ownership of configuration, curricular grounding, deployment, privacy defaults, and evaluation. Pedagogical alignment thus becomes a configuration and authoring problem owned by teachers, not a model-training problem owned by ML engineers.

The paper is organized around three research questions, which map onto its three contributions: **RQ1**: What do students and teachers at a technically oriented secondary school require of an institutional AI tutor? (an initial requirements evidence base, traced explicitly to design decisions); **RQ2**: How can a school align tutors with those requirements through teacher-owned configuration rather than model training? (a teacher-configurable architecture of reusable, teacher-authored modules served over in-region, open-weight models); **RQ3**: Can a small, in-region open-weight model deliver the intended Socratic tutoring behavior, and how much prompt detail must teachers provide? (a reproducible evaluation path with preliminary results).

2 Related Work and Positioning

Recent work converges on one idea, *pedagogical alignment*: a general-purpose LLM, used as-is, behaves as a generic information provider that hands over answers, whereas effective teaching requires scaffolded, Socratic guidance [7]; the works differ in *how* they close that gap. One family aligns models through training: Sonkar et al. [7] use preference learning to make a model guide rather than answer, and contribute metrics that quantify scaffolding versus direct answers. A second family aligns through prompt or configuration: Gabrovšek and Rihtaršič [2] deploy a configurable generative-AI tutor in an undergraduate STEM lab and report that structured, declarative prompt styles were rated highest by students. The broader effectiveness of (non-LLM) intelligent tutoring systems is well established, with meta-analytic gains around 0.66 SD over conventional instruction [4].

Novedu adopts the configuration route, the most realistic lever available to a secondary school, which cannot curate preference datasets or fine-tune models but can author prompts and use in-region open-weight models. Our contribution is largely orthogonal: where this literature studies *the tutor*, we study *the school's path to fielding tutors*, an institutional embedding (teacher ownership, curricular fit, privacy, in-region inference, evaluation) that international guidance increasingly argues schools must address as a whole. For evaluation we build on work measuring the *pedagogical* quality of responses rather than answer correctness [8] and on the LLM-as-a-judge paradigm and its biases [9].

3 Requirements and Design Process

From exploratory use to a scope-reduced platform. Novedu emerged from a requirements-driven progression rather than an up-front specification: a vision document committed to teacher-edited prompts, per-tutor model choice, open-source components, and self-hostable hosting. Students then composed existing open-source products (Open WebUI with a model gateway, school single sign-on, per-user budgets) into a working environment (Fig. 1a); this first version leaned on retrieval-augmented generation (RAG) [5] to bring curriculum material into the tutor, and served as an instrument for learning where a generic chatbot UI falls short pedagogically; rather than being evaluated as a product in isolation, its use informed the requirements process and motivated the scope reduction reported here. Guided by a senior product owner (the head of the computer-science department), the team produced a requirements specification and a full user-experience (UX) prototype of an institutional platform; a cross-disciplinary teacher panel received it well but judged it too large to deploy quickly, prompting a scoping decision: reduce the vision to a minimal deployable tutor workflow, deferring institutional features such as admin consoles and user/group management. Across the project, students built the Open WebUI prototype, the requirements and UX artifacts, and parts of the continuation; the final minimum viable product (MVP) seed was created in a teacher-led workshop and continued cooperatively.

Requirements survey. Alongside this process, a two-variant questionnaire (students and teachers), fielded in supervised in-class sessions as a pretest, yielded 318 valid responses (284 students, 34 teachers) on a 4-point Likert scale with no neutral midpoint (theoretical mean 2.5). The most important caveat is sample bias: the school is informatics-heavy (72.9% of student respondents in the IT branch), so the findings do not generalize directly to general-education schools. Four mutually reinforcing requirements dominate across both groups. *Curricular grounding* is the strongest single signal: the top student item is that the tutor know their concrete lesson content ($M = 3.53$). *Factual reliability* is critical on both sides: wrong answers are students' most frequent objection and teachers' top property (84.8%). *Configurable scaffolding* beats any fixed mode: students' top explanation choice is "it depends, let me choose" (52%). *Non-surveillance oversight* is required by both: students dislike teachers reading even anonymized chats (57%), while teachers want aggregated statistics (75.8%) over individual chats (6%). Two adoption findings frame the rest: teachers are willing but underprepared (only 6% feel well-informed on the legal framework; training demand $M = 3.27$), and willingness to author tutors is conditional on low effort.

Table 1. Requirements-to-design traceability. Survey signals (Chapter 3) drive design responses (Chapter 4); the gaps that remain are discussed in Chapter 6.

Survey signal	Design implication	Novedu response
Tutor must know lesson content (M=3.53)	Curricular grounding > generic chat	Teacher-authored topic / tutor modules
Wrong answers top objection	Reliability must be evaluated	SocraticTutorEval harness (Chap. 5)
Want configurable help style (52%)	One Socratic mode is too rigid	Reusable YAML behavior fragments
Reject read-along; want aggregate stats	Oversight must be non-surveillance	Anonymous defaults, aggregate code stats
Want training and templates	Adoption depends on low effort	Prompt library, starter tutors, AI authoring aids

Table 1 traces each requirement to a design response in Novedu, making the design decisions answerable to the evidence. Chapter 4 details these responses; Chapter 6 discusses the gaps that remain.

4 Novedu Design

4.1 Teacher-Configurable Tutors as Modular Configuration

A single idea organizes the architecture: a tutor is a plain, human-readable file, and the application is a configuration-driven layer around it, so pedagogy lives in teacher-authored configuration, not in application code or model weights. Two kinds of configuration files, written in the human-readable YAML format, are composed. *Fragment libraries* are centrally maintained, reusable prompt modules (a persona, ground rules such as a "never give away the solution" Socratic policy, a topic description), each versioned and parameterized by an input schema. *Tutor files*, written per subject, select fragments, fill in their inputs, add final instructions, and reference a model. A teacher therefore writes only a short, structured definition: a complete tutor file fits on a dozen lines naming a model, referencing fragments (for example a Socratic no-solution policy and a topic), and adding final instructions; full examples are in the project repository. At request time a framework-agnostic pipeline fetches, validates, and assembles the fragments into a system prompt for a single agent answering over an in-region open-weight model. Unlike the exploratory prototype, the coded MVP carries no retrieval pipeline: curriculum context lives in teacher-authored fragments rather than in RAG, trading automatic recall for teacher control and a simpler, self-hostable deployment.

This pedagogical content is produced by a cross-disciplinary, teacher-owned effort: a shared, version-controlled library of tutor fragments authored by teachers of German, English, mathematics, biomedicine, and programming, so the same behavior module serves all subjects alike. It is the clearest single artifact

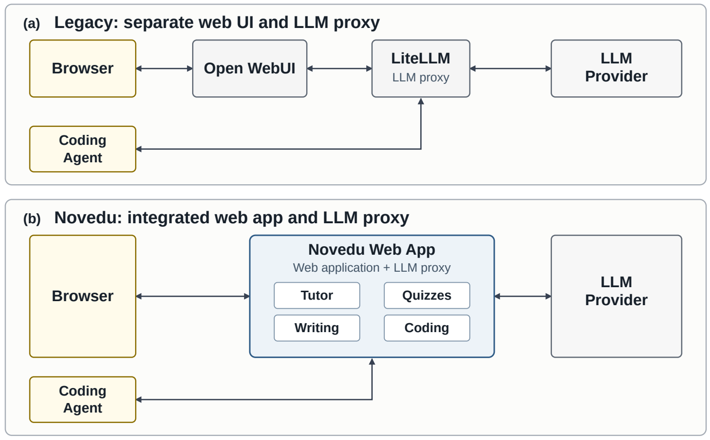

Fig. 1. From composed products to an integrated platform: (a) the exploratory prototype; (b) the Novedu MVP serving the four activity types (Chapter 4.2).

of institutional embedding: pedagogy shaped by teachers across faculties, not by the development team alone.

Authoring usability. A first presentation to non-technical teachers showed that authoring YAML by hand is a real usability barrier, matching the survey's low-effort precondition. We address this without abandoning the plain-file model: a JSON Schema gives autocompletion and inline validation in ordinary editors, an AI authoring aid turns prose into a valid tutor definition, and a command-line verification tool returns precise errors. A graphical editor that hides YAML entirely is the next step (a final-year student project).

4.2 One Abstraction, Several Activities

The configurable-file idea generalizes beyond chat: a teacher mints a short *code* pointing at an activity definition, and one access-gating and statistics mechanism serves whichever activity the code names. Fig. 1 contrasts the resulting architecture with the exploratory prototype: where the legacy stack composed a generic chat front end (Open WebUI) with a separate LLM proxy (LiteLLM), Novedu integrates web application and LLM proxy into one teacher-configurable app serving all four activity types. Tutors carry the most pedagogical weight (a Socratic-by-default persona, rules, topic, and model); quizzes pair open-ended questions with a server-only grading rubric scored by a stateless grader; writing provides a split-screen coach that can read but not edit the student's draft; and coding injects a teacher-authored system prompt into an OpenAI-compatible proxy for students' coding agents. Anonymity is the default except where graded writing requires attribution.

4.3 Self-Hosting and Privacy Defaults

Institutional fit rests on a few deployment decisions. Inference avoids commercial cloud LLMs: open-weight, quantized models (roughly 27–31 billion parameters) are served behind an OpenAI-compatible API, addressed server-side only, by an in-region GPU server operated by the Software Competence Center Hagenberg (SCCH), an Austrian research center, through its general-purpose, open-source, self-hostable *LLM:2go* offering, made available to the school at no cost through a government-funded research project. Identity is the school's own (single sign-on, teacher actions gated by group membership), and privacy is the default: activities are anonymous unless a teacher opts out, the quiz grader stores nothing, telemetry carries no message content, and chat history is not retained. These choices support GDPR- and AI-Act-aware deployment by avoiding commercial, data-collecting LLM services and keeping processing in-region; SCCH is nonetheless a separate processor, and a full compliance assessment is outside this paper's scope. Building the platform is itself a cooperative student software-engineering project under teacher direction, a secondary computing-education contribution we do not develop here.

5 Preliminary Evaluation

The survey measures stated preferences. To measure *behavior*, whether a configured tutor actually withholds the answer, we built *SocraticTutorEval*: given a partial student–tutor conversation ending on a student turn, it generates the next tutor reply with several LLMs and has an LLM judge score each against a per-scenario rubric (a pass requires no negative-criterion violation and $\geq 70\%$ of positive criteria met). A failing reply does not indicate a wrong answer; it indicates a departure from the behavior the teacher specified, primarily abandoning the Socratic stance by volunteering the solution. We use it to probe RQ3 on one programming topic (linked lists): three system prompts of increasing length for the same tutor (a 6-line *minimal* prompt of two sentences, an 84-line *short*, and a 172-line *full* prompt), across one in-region open-weight model (the SCCH endpoint) and four frontier baselines (Table 2). Each cell aggregates 36 judged conversations (12 scenarios $\times$ 3 replies), each scored once by `gpt-5.4` at high reasoning effort; rates should be read descriptively, as one pass shifts a cell by 2.8 points.

Two findings stand out. First, *prompt detail matters substantially* (RQ2): moving from the full prompt to the two-sentence minimal prompt lowers the pass rate for every model, by 14 points for the in-region model and up to 33 points for `gpt-5.4-mini`, reinforcing the need for authoring support (Chapter 4.1). Second, on this Socratic-adherence metric a small *in-region open-weight* model is viable (RQ3): `gemma-4-31B` matches the best frontier configuration on the full prompt (61.1%) and is descriptively the most robust to a terse prompt (47.2%, versus $\leq 33.3\%$); higher reasoning effort did not help. The frontier models do not score lower because they are weaker: their characteristic failure is over-responding, handing the student a full answer and breaking the Socratic

Table 2. Pass rate (%) by model and prompt length on the linked-lists scenarios.

Model	**6-line**	**84-line**	**172-line**
gemma-4-31B-FP8 (SCCH, in-region)	**47.2**	55.6	**61.1**
gpt-oss-120b (open weight, medium)	33.3	52.8	61.1
gpt-5.4-mini	27.8	55.6	61.1
gpt-5.4 (medium reasoning)	33.3	47.2	55.6
gpt-5.4 (high reasoning)	27.8	50.0	47.2

constraint, and it proved easier to steer a small model into Socratic behavior than to restrain a large one from over-helping. These preliminary results make in-region open-weight inference plausible for further classroom testing without establishing general superiority over commercial models. Notably, these prompts predate the fragment model; maintaining a long prompt by hand motivated the reusable-fragment design (Chapter 4.1).

These results are preliminary: a single topic, small per-cell counts, and an LLM judge whose biases and calibration require scrutiny [9] and may diverge from human teachers on helpfulness [8]. The harness has been shared with the teacher group and first cross-teacher experiments have begun; validating the judge against independent teacher ratings is part of the ongoing, multi-subject analysis. Model IDs, decoding parameters, the judge, and prompts are archived with the harness for reproducibility.

6 Limitations and Next Steps

The platform is a learning prototype: its interface is functional rather than polished, it is not yet usable on smartphones (a real gap for students), and it is single-tenant. The evaluation is correspondingly bounded: the survey is a single-site, supervised, self-report pretest, with an informatics-heavy sample that limits external validity; the harness covers one topic and depends on an LLM judge that needs calibration; and we have no classroom learning-outcome data yet. Our own survey finding that students reject teacher read-along on individual chats even when anonymized (57%) makes a consented, individual-level log analysis as in [2] hard to justify; future log-based work must run on aggregated data. We will broaden the questionnaire to additional schools in the coming academic year to remove the branch bias of the pretest.

The most important gap is that our strongest requirement outruns the implementation: students demand that the tutor know their concrete lesson material, yet the MVP supplies this only through teacher-authored prompt modules, not retrieval over the school's learning-management system. Closing that gap is the priority next step, but not necessarily by defaulting to RAG: we plan to test whether teacher-authored prompts, optionally extended with a lightweight tool-based (grep-style) lookup over curriculum material, are competitive with a full RAG pipeline, since recent evidence finds no universal winner between retrieval and long-context prompting [6] and documents characteristic RAG failure modes [1]. Equally central is adoption: a teacher-configurable platform works only if teachers can and will configure it, so the graphical authoring editor and a teacher-training effort (the survey's clear demand) are conditions for success. Further work includes a behavioral, in-classroom evaluation on aggregated, anonymized data and shipping the coding proxy; the cooperative-development model and the configuration-over-training architecture may then transfer to other computer-science schools.

7 Conclusion

Novedu is a design case of how a school can move from exploratory AI use toward a teacher-configurable, privacy-aware tutoring infrastructure. The survey answers RQ1 with four requirements: curricular grounding, factual reliability, configurable scaffolding, and aggregated oversight. The architecture answers RQ2 by making alignment a teacher-owned configuration problem rather than a model-training one: tutors are assembled from reusable, plain-text modules and served over in-region, open-weight models with anonymous defaults. The evaluation harness answers RQ3 tentatively: a small in-region model held the intended Socratic stance as reliably as frontier baselines, provided teachers supply detailed prompts. The platform and its evaluation remain ongoing.

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